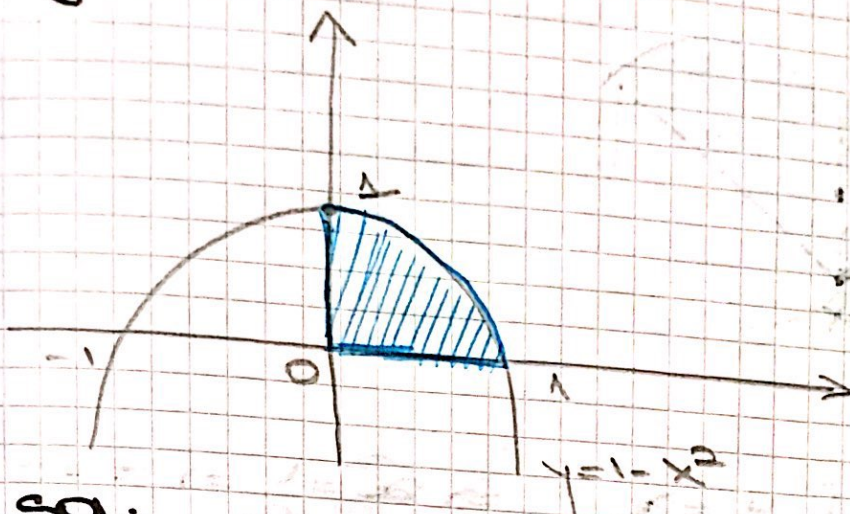


$$A = \{ (x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, 0 \leq y \leq 1-x^2 \}$$
Sol:

Def 2:  $[0,1] \rightarrow \mathbb{R}, f(x) = 1 - x^2$  cond  $\Rightarrow$

$\Rightarrow$  und  $(R) \in [0,1] \Rightarrow$

$\pi f \in \mathcal{H}(\pi\mathbb{R})$  &  $\lambda(\pi f) = \int_0^1 f(x) dx$   
 Obs:  $A = \pi_0 \Rightarrow$

Obs :  $A = \mathbb{F}_p$

$$\Rightarrow \lambda(x) = \int_0^x f(x) dx = \int_0^x (1-x^2) dx =$$

81  
↑  
vac  
dipole

$$B = \{ (x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 4 \} \in \mathbb{R}^2$$

$$B \in \mathcal{I}(\mathbb{R}^2)$$

উঃ

Def  $f_1: [-2, 2] \rightarrow \mathbb{R}, f_1(x) = \sqrt{4-x^2}$

$$f_2: [-2, 2] \rightarrow \mathbb{R}, f_2(x) = -\sqrt{4-x^2}$$

$f_1, f_2$  sind  $(\pi) = G \circ \pi_1, G \circ \pi_2 \in J(\mathbb{R}^2)$

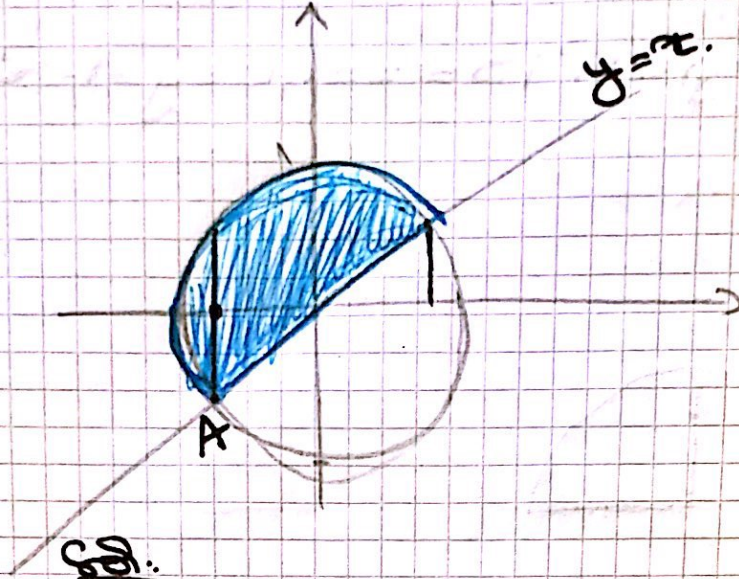
$$\lambda(Gz_1) = \lambda(Gz_2) = 0.$$

$$\text{Ded } B = \{ \neg \varphi_1 \cup \neg \varphi_2 \rightarrow B \in J(\mathcal{A}) \}$$

$$\Rightarrow \lambda(B) \leq \lambda(G \setminus x_1) + \lambda(G \setminus x_2) = 0 \Rightarrow$$

$$\Rightarrow \chi(B) = 0.$$

9)  $C = \{ (x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1, y \geq x \}$



find

$$A: \begin{cases} y = x \\ x^2 + y^2 = 1 \end{cases} \Rightarrow x^2 = \frac{1}{2} \Rightarrow x = \pm \frac{1}{\sqrt{2}}$$

$$x = -\frac{1}{\sqrt{2}} \Rightarrow \left( -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)$$

Define  $f_1: [-1, 1] \rightarrow \mathbb{R}, f_1(x) = \sqrt{1-x^2}$

$f_2: [-1, \frac{1}{\sqrt{2}}] \rightarrow \mathbb{R}, f_2(x) = -\sqrt{1-x^2}$

$f_3: [-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}] \rightarrow \mathbb{R}, f_3(x) = x$

$f_1, f_2, f_3$  cont. on  $D_f \Rightarrow \text{int}(R) =$

$$\Rightarrow C_1 = \bigcup_{f_1, f_2} [-1, -\frac{1}{\sqrt{2}}] \rightarrow \mathbb{R} \in \mathcal{I}(\mathbb{R}^2)$$

$$C_2 = \bigcup_{f_3} [-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}] \rightarrow \mathbb{R}$$

(P3)  $\Rightarrow \lambda(C_1) = \int_{-1}^{-\frac{1}{\sqrt{2}}} (\sqrt{1-x^2} - (-\sqrt{1-x^2})) dx +$

$$\lambda(C_2) = \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} (\sqrt{1-x^2} - x) dx$$

$$\Rightarrow \lambda(C) = \lambda(C_1) + \lambda(C_2)$$

Sol 2:

Pravind mäsuvabli-  
tatea Jordan no  
arie, dăvând

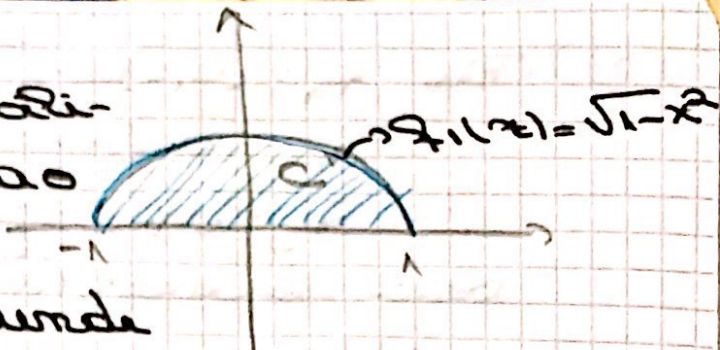
că  $\lambda(C) = \lambda(C')$  unde

$$C' = \{(x, y) \mid x^2 + y^2 \leq 1, y \geq 0\}$$

Deci  $C' = \pi_f$ , unde  $f: [-1, 1] \rightarrow \mathbb{R}$

$$f(x) = \sqrt{1-x^2} \text{ cont}$$

$$\begin{aligned} \text{pe } [-1, 1] \Rightarrow C' \in \mathcal{J}(\mathbb{R}^2) \text{ și } \lambda(C') &= \lambda(\pi_f) = \\ &= \int_{-1}^1 \sqrt{1-x^2} dx = \\ &= \dots \end{aligned}$$



b)  $D = \{(x, y) \in \mathbb{R}^2 \mid |x| \leq 1, |y| = x^2\} \in \mathcal{J}(\mathbb{R}^2)$   
 $-1 \leq x \leq 1, -x^2 \leq y \leq x^2$

Sol:

Fie  $f, g: [-1, 1] \rightarrow \mathbb{R}$

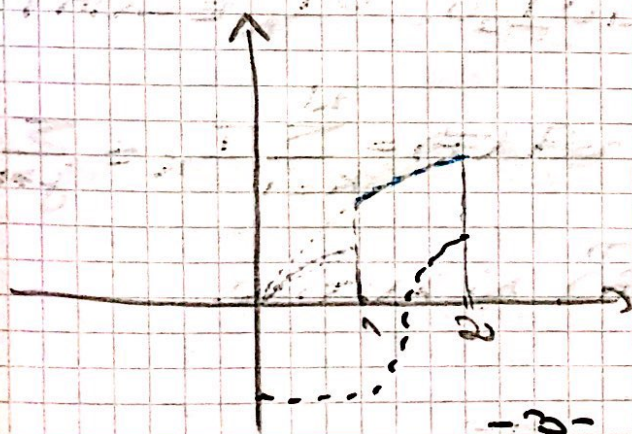
$$f(x) = -x^2 \mid \text{cont} \Rightarrow \text{unit}(\mathbb{R}) \Rightarrow$$

$$g(x) = x^2$$

$$\Rightarrow D = \pi_{f,g} \in \mathcal{J}(\mathbb{R}^2) \text{ și}$$

$$\begin{aligned} \lambda(D) &= \lambda(\pi_{f,g}) = \int_{-1}^1 (x^2 + x^2) dx = \\ &= 2x^3 \Big|_{-1}^1 = \frac{4}{3} \end{aligned}$$

ii)  $f: [0, 2] \rightarrow \mathbb{R}, f(x) = \begin{cases} \sin x, & x \in \mathbb{Q}, x \leq 1 \\ x, & x \in \mathbb{Q}, x > 1 \\ -\cos x, & x \in (\mathbb{R} \setminus \mathbb{Q}) \end{cases}$



Fix  $f_1: [0,2] \rightarrow \mathbb{R}, f_1(x) = \sin \pi x$

$f_2: [0,2] \rightarrow \mathbb{R}, f_2(x) = x$

$f_3: [0,2] \rightarrow \mathbb{R}, f_3(x) = \cos \pi x$

const  $\Rightarrow$  int  $(f) \Rightarrow G_{f_1}, G_{f_2}, G_{f_3}$

$$\begin{aligned} \text{si } \lambda(G_{f_1}) &= \lambda(G_{f_2}) = \lambda(G_{f_3}) = 0 \\ \Rightarrow 0 &= \lambda(G_{f_1} \cup G_{f_2} \cup G_{f_3}) = \\ &= \lambda(G_{f_1}) + \lambda(G_{f_2}) + \lambda(G_{f_3}) = 0 \end{aligned}$$

$$\Rightarrow H = G_{f_1} \cup G_{f_2} \cup G_{f_3} \in \mathcal{I}(\mathbb{R}^2) \quad \lambda(H) = 0.$$

for  $G_{f_1} \subseteq H$

$$\Rightarrow G_{f_1} \in \mathcal{I}(\mathbb{R}^2) \text{ si } \lambda(G_{f_1}) = 0$$

**Ex 1)** Fix  $A \subseteq \mathbb{R}^m$  compact, negligible  
 Lebesgue. Atunci  $A \in \mathcal{I}(\mathbb{R}^m)$  si  $\lambda(A) = 0$ .

Sol:

Fix  $\varepsilon > 0$  fixat.

• A negligible  $(\varepsilon) \Rightarrow \exists (I_j)_{j \geq 1}$  un sir  
 de intervale deschise cat  $A \subset \bigcup_{j=1}^{\infty} I_j$   
 si  $\sum_{j=1}^{\infty} \text{vol}(I_j) < \varepsilon$

• A compact  $\Rightarrow \exists N \in \mathbb{N}$  cat  
 $A \subset I_1 \cup I_2 \cup \dots \cup I_N$ .

Fix  $\bar{I} = I_1 \cup I_2 \cup \dots \cup I_N \in \mathcal{E}(\mathbb{R}^m)$  si  
 $\bar{I} \supset A$ .

$$\text{for } \lambda(\bar{I}) = \text{vol}(\bar{I}) = \sum_{k=1}^N \text{vol}(I_k) \leq \sum_{j=1}^{\infty} \text{vol}(I_j)$$

$\leq \varepsilon \Rightarrow$

$$A \in \mathcal{I}(\mathbb{R}^m) \text{ si } \lambda(A) = 0.$$

Ex2

Dacă  $A \in \mathcal{J}(\mathbb{R}^n)$  cu  $\lambda(A) = 0$ , arătați că  $A$  este neglijabil.

Sol:

Fie  $\varepsilon > 0$  fixat

$A \in \mathcal{J}(\mathbb{R}^n)$  cu  $\lambda(A) = 0 = \lambda(A) = 0$

$\Rightarrow \exists F \in \mathcal{E}(\mathbb{R}^n)$  cu  $A \subset F$  și  $\lambda(F) = \text{vol}(F) < \varepsilon$

Deci  $F \in \mathcal{E}(\mathbb{R}^n) \xrightarrow{\text{def}} \exists F_1, F_2, \dots, F_p$

$\exists I_1, I_2, \dots, I_p$  intervale cu intersecții disjuncte două câte două și

$$F = \bigcup_{j=1}^p F_j$$

Fie  $(I_k)_{k=1}^{\infty}$  cu  $I_k = \emptyset, \forall k \geq p$ .

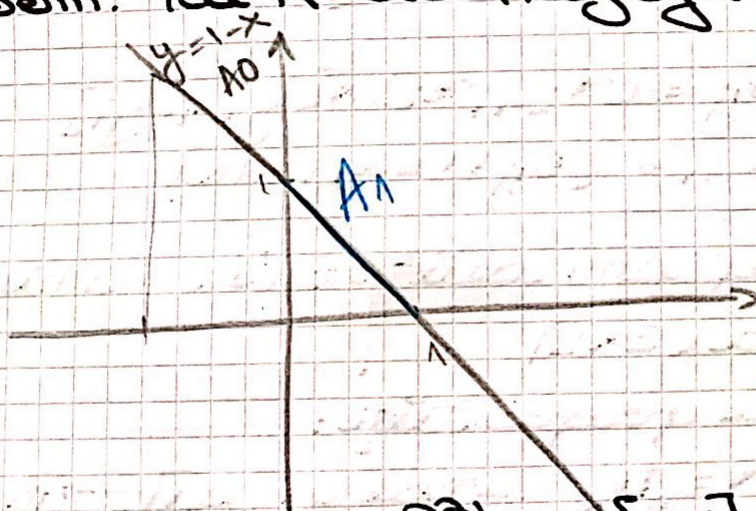
Atunci  $A \subseteq \bigcup_{k=1}^{\infty} I_k$  și

$$\sum_{k=1}^{\infty} \text{vol}(I_k) = \sum_{k=1}^{\infty} \text{vol}(I_k) = \lambda(F) < \varepsilon \Rightarrow$$

$\Rightarrow A$  neglijabil.

Ex3

Fie  $A = \{(x, y) \in \mathbb{R}^2 \mid x + y - 1 = 0\}$   
 Dem. că  $A$  este neglijabil.



$$A_1 = \{(x, y) \in \mathbb{R}^2 \mid x \in [0, 1], y = 1 - x\}$$

$A_1 \in \mathcal{J}(\mathbb{R}^2), \lambda(A_1) = 0$ .

Fie  $\varphi: [0, 1] \rightarrow \mathbb{R}, \varphi(x) = 1 - x$  cont.  $\Rightarrow$   
 $\Rightarrow$  unit.  $\Rightarrow A_1 = G \circ \varphi \in \mathcal{J}(\mathbb{R}^2)$  și

$$\lambda(A_1) = \lambda(G \circ \varphi) = 0 \Rightarrow A \text{ neglijabil}$$

Fie  $A_m = \{ (x, y) \in \mathbb{R}^2 \mid x \in [m-1, m], y = 1-x \}$ ,  $\forall m \in \mathbb{Z}$

Analog ca mai sus,  $A_m \in \mathcal{J}(\mathbb{R}^2)$  cu  $\lambda(A_m) = 0$ ,  $\forall m \in \mathbb{Z} \Rightarrow A_m$

Obs că  $A = \bigcup_{m \in \mathbb{Z}} A_m$

$\Rightarrow A$  sumă numărabilă de  
multimi măsurabile  $\Rightarrow$   
 $\Rightarrow A$  măsurabilă

**Ex 4**

Fie  $A = \{ (x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, 0 \leq y \leq x \}$

a) Dem. că  $A \in \mathcal{J}(\mathbb{R}^2)$  (cu def)

b) Folosind def. integralei Riemann, arătați că  $\int_A 1$

$f: [0, 1] \times [0, 1] \rightarrow \mathbb{R}$

$f(x, y) = \begin{cases} 1, & (x, y) \in A \\ 0, & (x, y) \in I \setminus A \end{cases}$  este

int. (R) și că  $\iint_I f(x, y) dx dy$

a)  $D_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j]$   
 $\Delta_m = \{ D_{ij} \mid 1 \leq i, j \leq m \}$

Cu def am arătat în seminara anterioară (Ex 1)

Obs: Cu proprietăți:

$A = \pi_{BC} \pi_{AD}$  unde  $g: [0, 1] \rightarrow \mathbb{R}$ ,  $g(x) = x$

$\Rightarrow A \in \mathcal{J}(\mathbb{R}^2)$  cu  $\lambda(A) = \int_0^1 x dx$

$= \frac{1}{2}$

$$S_f(\Delta_m) = \sum_{i,j=1}^m m_{ij} \phi(D_{ij})$$

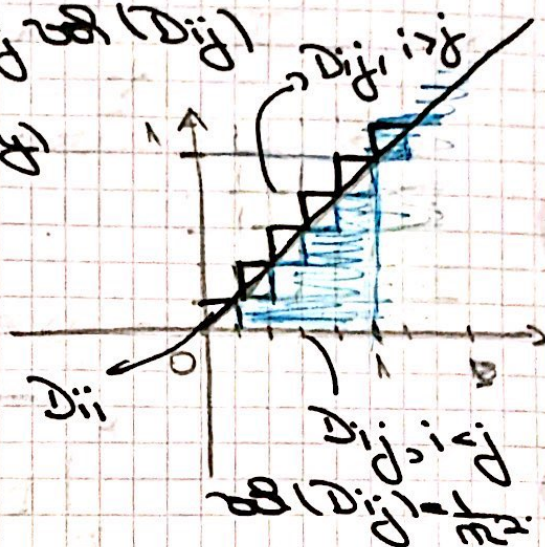
$$m_{ij} = \sum_{(x,y) \in D_{ij}} \phi(x,y)$$

$$S_f(\Delta_m) = \sum_{i,j=1}^m \phi_{ij} \phi(D_{ij})$$

$$M_{ij} = \sup_{(x,y) \in D_{ij}} \phi(x,y)$$

$$m_{ij} = \begin{cases} 1, & j < i \\ 0, & j \geq i \end{cases}$$

$$M_{ij} = \begin{cases} 1, & j \leq i \\ 0, & j > i \end{cases}$$



$$S_f(\Delta_m) = \sum_{1 \leq j < i \leq m} 1 \cdot \phi(D_{ij}) = \sum_{1 \leq j < i \leq m} \frac{1}{2^2}$$

$$= (m-1) + (m-2) + \dots + 2 + 1 \cdot \frac{1}{2^2} =$$

$$= \frac{m(m-1)}{2} \cdot \frac{1}{2^2} = \frac{m-1}{2} = \frac{1}{2} \left( 1 - \frac{1}{2^2} \right)$$

$$S_f(\Delta_m) = \sum_{1 \leq j \leq i \leq m} 1 \cdot \phi(D_{ij}) = \sum_{1 \leq j \leq i \leq m} \frac{1}{2^2}$$

$$= (m + m-1 + \dots + 2 + 1) \cdot \frac{1}{2^2} =$$

$$= \frac{m(m+1)}{2} \cdot \frac{1}{2^2} = \frac{m+1}{2} = \frac{1}{2} \left( 1 + \frac{1}{2^2} \right)$$

$$\lim_{m \rightarrow \infty} S_f(\Delta_m) = \lim_{m \rightarrow \infty} \frac{1}{2} \left( 1 - \frac{1}{2^2} \right) = \frac{1}{2}$$

$$= \lim_{m \rightarrow \infty} S_f(\Delta_m) = \lim_{m \rightarrow \infty} \frac{1}{2} \left( 1 + \frac{1}{2^2} \right) = \frac{1}{2}$$

$$= \lim_{m \rightarrow \infty} S_f(\Delta_m) = \frac{1}{2}$$

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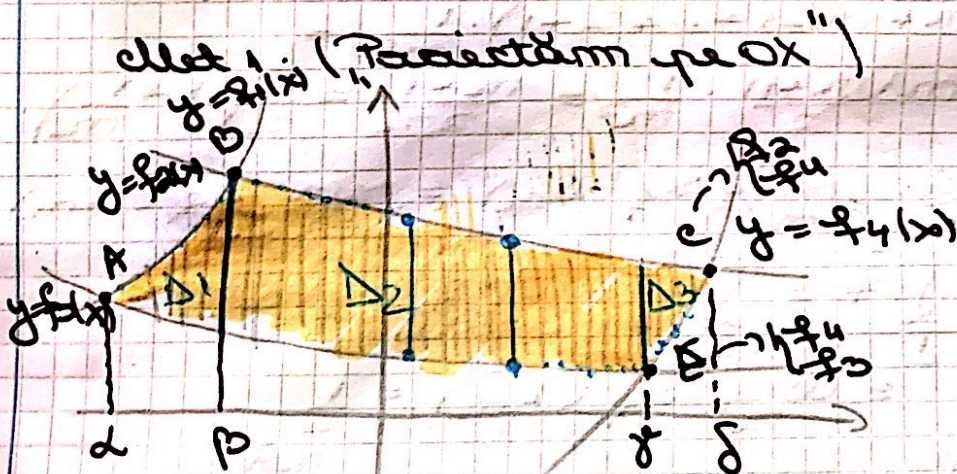
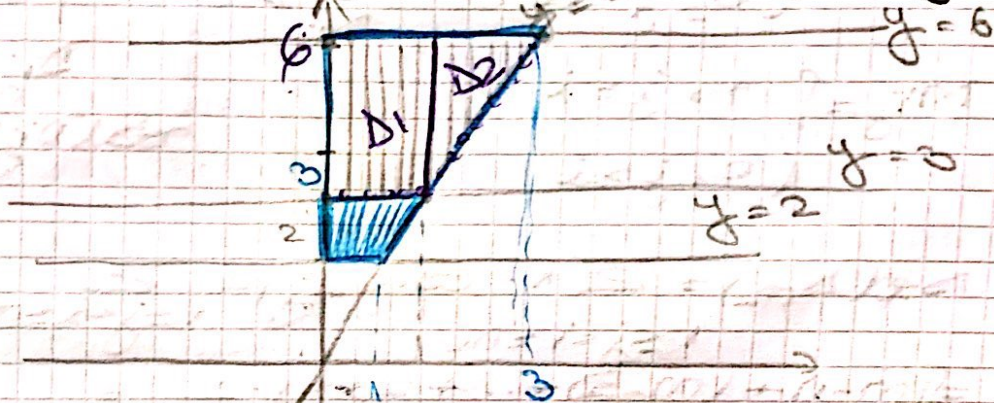
# Integrale duble. Metode colub

## I. "Prin tăieturi"

Ex1:  $I = \iint_D (x+4y) dx dy$  unde

-  $D$  delimitat de:  $\begin{cases} y=2x \\ y=2 \\ y=6 \\ x=0 \end{cases}$

-  $D = \{(x,y) \in \mathbb{R}^2 \mid y \geq 2x, y \geq 2, y \leq 6, x \geq 0\}$



$\iint_D f(x,y) dx dy$   $D = D_1 \cup D_2 \cup D_3$   
 ①  $f \in \mathcal{F}(\mathbb{R}^2)$

$$D_1: \alpha \leq x \leq \beta$$

$$f_3(x) \leq y \leq f_1(x)$$

cont pe  $[\alpha, \beta]$   
proiectia pe OX

$$D_2: \beta \leq x \leq \gamma$$

$$f_3(x) \leq y \leq f_2(x)$$

cont pe  $[\beta, \gamma]$

$$D_3: \gamma \leq x \leq \delta$$

$$f_4(x) \leq y \leq f_2(x)$$

cont pe  $[\gamma, \delta]$   
"Adapmantul de

introducerea: pe ce ecuatii  
mai putem?"

$$\iint_D f = \iint_{D_1} f + \iint_{D_2} f + \iint_{D_3} f =$$

$$= \int_{\alpha}^{\beta} \left( \int_{f_3(x)}^{f_1(x)} f(x, y) dy \right) dx +$$

$$+ \int_{\beta}^{\gamma} \left( \int_{f_3(x)}^{f_2(x)} f(x, y) dy \right) dx + \int_{\gamma}^{\delta} \left( \int_{f_3(x)}^{f_2(x)} f(x, y) dy \right) dx$$

Avem acum la ma:

$$D = D_1 \cup D_2$$

$$D_1: 0 \leq x \leq 1$$

$$2 \leq y \leq 6$$

constanta pe  
deputa ingri  
cont pe  $[0, 1]$

$$D_2: 1 \leq x \leq 3$$

$$2x \leq y \leq 6$$

cont pe  $[1, 3]$

SAU

$$D_1 = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1; 2 \leq y \leq 6\}$$

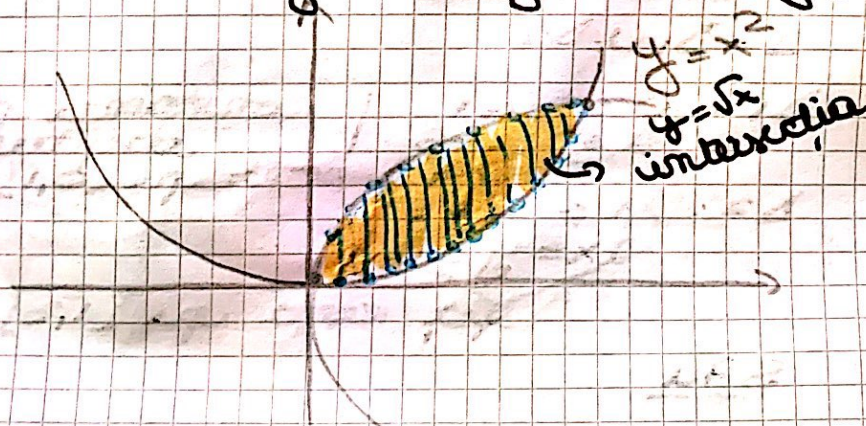
$$D_2 = \{(x, y) \in \mathbb{R}^2 \mid 1 \leq x \leq 3; 2x \leq y \leq 6\}$$

Exid  $D_1, D_2 \in \mathcal{J}(\mathbb{R}^2)$  si  $f: D \rightarrow \mathbb{R}$ ,  
 $f(x, y) = x + 4y$  cont pe  $D \Rightarrow \text{int}(A)$   
 pe  $D$  (analog pe  $D_1, D_2$ ).

$$\begin{aligned}
 I &= \iint_D f = \iint_{D_1} f + \iint_{D_2} f = \\
 &= \int_0^1 \left( \int_2^6 (x+4y) dy \right) dx + \int_1^3 \left( \int_{2x}^6 (x+4y) dy \right) dx \\
 &= \int_0^1 (xy + 4y^2) \Big|_2^6 dx + \int_1^3 (xy + 4y^2) \Big|_{2x}^6 dx \\
 &= \int_0^1 (x \cdot 6 + 2 \cdot 6^2 - (x \cdot 2 + 2 \cdot 2^2)) dx + \\
 &\quad + \int_1^3 ((x \cdot 6 + 2 \cdot 6^2) - (x \cdot 2x + 2 \cdot (2x)^2)) dx \\
 &= \int_0^1 (4x + 64) dx + \int_1^3 (6x + 72 - 2x^2 - 8x^2) dx \\
 &= \dots
 \end{aligned}$$

Ex2.  $I = \iint_D (x + xy - 2) dx dy$

$$D = \{(x, y) \in \mathbb{R}^2 \mid y \geq x^2, x \geq y^2\}$$



$$\begin{aligned}
 \begin{cases} y = x^2 \\ x = y^2 \end{cases} &\Rightarrow y = y^4 \Rightarrow y(y^3 - 1) = 0 \Rightarrow y = 0 \Rightarrow (0, 0) \\
 &\quad y = 1 \Rightarrow (1, 1)
 \end{aligned}$$

$$D: 0 \leq x \leq 1$$

$$x^2 = y = \sqrt{x}$$

cont. on  $[0,1]$ .

$$D \subset \mathbb{R}^2$$

$$f: D \rightarrow \mathbb{R}, f(x) = x + xy - 2 \text{ cont. on } D$$

$$\Rightarrow f \text{ int. (R) on } D \Rightarrow$$

$$\Rightarrow I = \int_0^1 \left( \int_{x^2}^{\sqrt{x}} (x + xy - 2) dy \right) dx =$$

$$= \int_0^1 \left( xy + \frac{xy^2}{2} - 2y \right) \Big|_{x^2}^{\sqrt{x}} dx =$$

$$= \int_0^1 \left( x\sqrt{x} + \frac{x\sqrt{x}^3}{2} - 2\sqrt{x} - x^3 - \frac{x^5}{2} + 2x^2 \right) dx$$

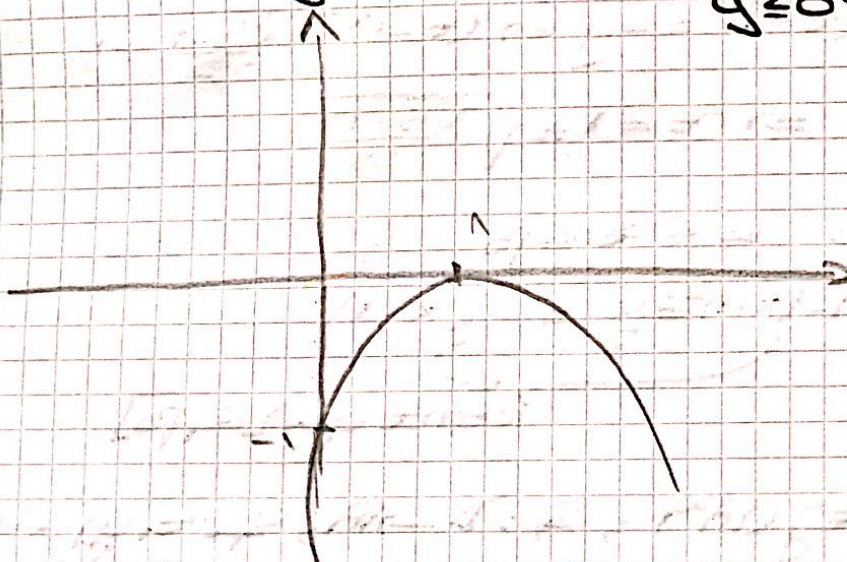
$$= x \cdot \frac{\sqrt{x}}{3/2} + \frac{1}{2} \cdot \frac{x^3}{3/2} - 2x \cdot \frac{\sqrt{x}}{3/2} - \frac{1}{4} \cdot \frac{x^4}{4} + \frac{2x^3}{3} \Big|_0^1$$

$$= \dots$$

Ex3

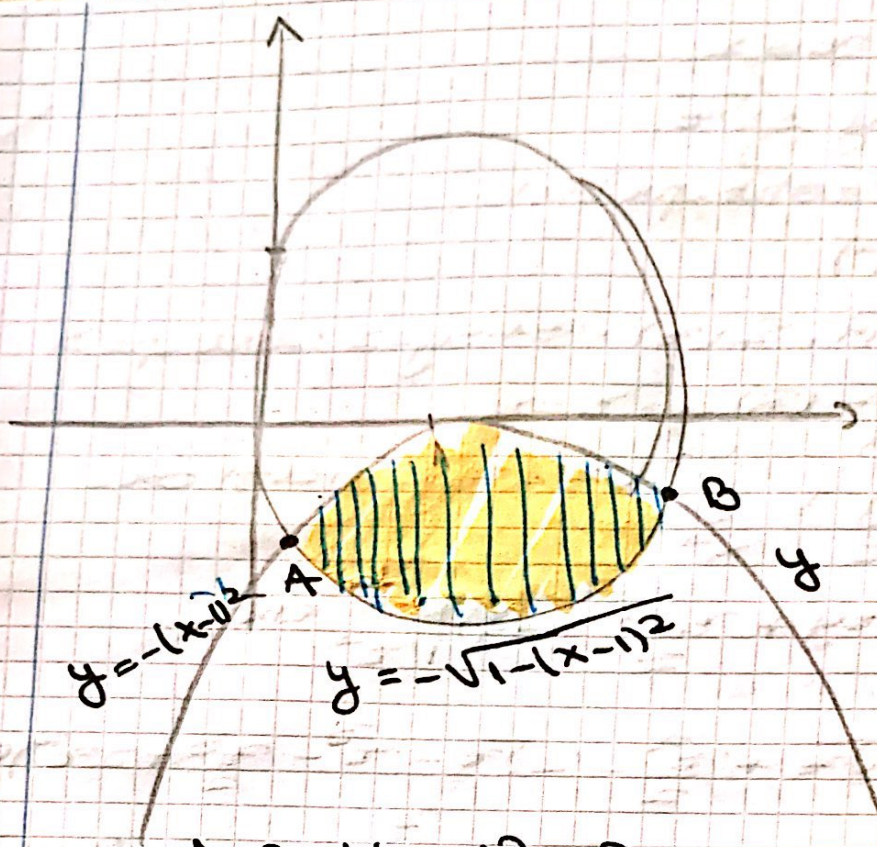
$$I = \iint_D 4xy \, dx \, dy$$

$$D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 2x, y \geq 0\} = \{(x, y) \mid (x-1)^2 + y^2 \leq 1, y \geq 0\}$$



$$x^2 + y^2 - 2x + 1 - 1 \leq 0$$

$$(x-1)^2 + y^2 \leq 1 \Rightarrow \mathcal{C}((1, 0), 1)$$



$$A, B: \begin{cases} (x-1)^2 + y^2 = 1 \\ y = -(x-1)^2 \end{cases}$$

$$\Rightarrow -y + y^2 = 1 \Rightarrow$$

$$\Rightarrow y^2 - y - 1 = 0$$

$$y_{1,2} = \frac{1 \pm \sqrt{5}}{2}$$

$$(x-1)^2 = -y \Rightarrow (x-1)^2 = \frac{\sqrt{5}-1}{2} \Rightarrow$$

$$\Rightarrow x = 1 \pm \sqrt{\frac{\sqrt{5}-1}{2}}$$

$$D: \alpha \leq x \leq \beta$$

$$-\sqrt{1-(x-1)^2} \leq y \leq -(x-1)^2$$

cont.  $y \in [\alpha, \beta]$

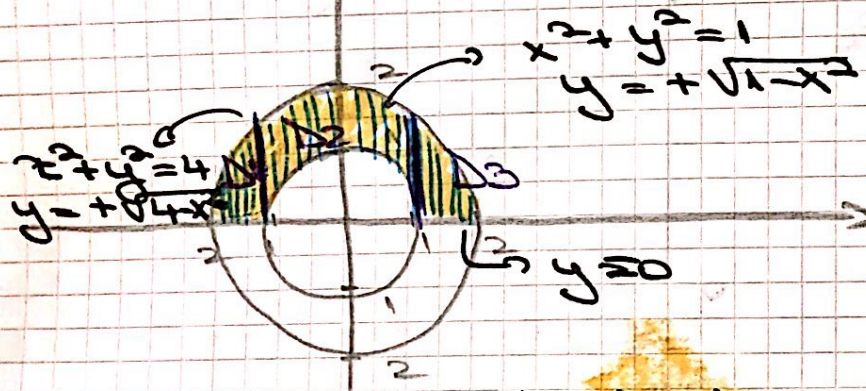
$\exists \varphi \in C(\mathbb{R}^2); \varphi: D \rightarrow \mathbb{R}, \varphi(x, y) = 4xy$   
 cont.  $y \in D \Rightarrow \text{int}(\varphi) \text{ on } D \Rightarrow$

$$\begin{aligned}
 &= I = \iint_D 2(x-y) dx dy = \\
 &= \int_{-2}^2 \left( \int_{-\sqrt{1-(x-1)^2}}^{-\sqrt{1-(x-1)^2}} 4xy dy \right) dx = \\
 &= \int_{-2}^2 2xy^2 \Big|_{-\sqrt{1-(x-1)^2}}^{-\sqrt{1-(x-1)^2}} dx = \\
 &= \int_{-2}^2 2x(x-1)^4 - 2x(1-(x-1)^2) dx = \dots
 \end{aligned}$$

$$\boxed{\text{Area}_D = \iint_D dx dy}$$

Exp: Find  $\iint_D$  calc area multiple

$$D = \{(x, y) \mid 1 \leq x^2 + y^2 \leq 4, y \geq 0\}$$



$$D = D_1 \cup D_2 \cup D_3, D_1, D_2, D_3 \in \mathcal{I}(\mathbb{R}^2)$$

$$f: D_i \rightarrow \mathbb{R}, f(x, y) = 1 \text{ for } i = 1, 2, 3$$

$$\Rightarrow f \text{ with } (R) \text{ on } D_i, i = 1, 2, 3$$

$$D_1: -2 \leq x \leq -1$$

$$D_2:$$

$$-1 \leq x \leq 1$$

$$D_3:$$

$$1 \leq x \leq 2$$

$$0 \leq y \leq \sqrt{4-x^2}$$

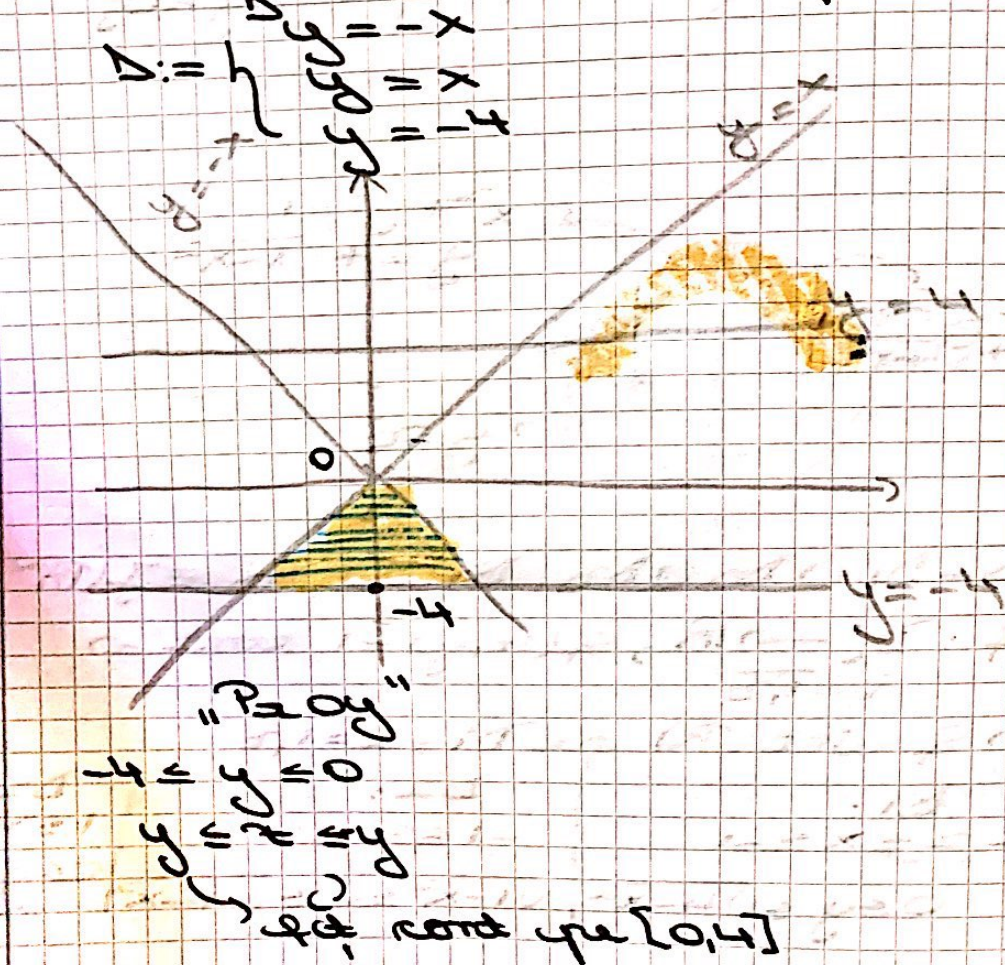
$$\sqrt{1-x^2} \leq y \leq \sqrt{4-x^2}$$

$$0 \leq y \leq \sqrt{4-x^2}$$

Geometric:  $\frac{\pi}{2}$

$$\begin{aligned}
 A_{\text{area}} &= \iint_{D_1} dx dy + \iint_{D_2} dx dy + \iint_{D_3} dx dy \\
 &= \int_{-1}^1 \left( \int_0^{\sqrt{4-x^2}} dy \right) dx + \int_{-1}^1 \left( \int_{\sqrt{1-x^2}}^{\sqrt{4-x^2}} dy \right) dx \\
 &\quad + \int_{-1}^1 \left( \int_0^{\sqrt{4-x^2}} dy \right) dx = \\
 &= \int_{-1}^1 \sqrt{4-x^2} dx + \int_{-1}^1 (\sqrt{4-x^2} - \sqrt{1-x^2}) dx \\
 &\quad + \int_{-1}^1 \sqrt{4-x^2} dx = \dots
 \end{aligned}$$

**Ex 4**  $I = \iint_D (x+2y) dx dy$  cont.  $\mu \Rightarrow \text{cont. } \mu \Rightarrow \text{cont. } \mu$



$$I = \int_{-1}^0 \left( \int_{-y}^y (x+3y) dx \right) dy =$$

$$= \int_{-1}^0 \left( \frac{x^2}{2} + 3xy \right) \Big|_{-y}^y dy =$$

$$= \int_{-1}^0 \left( \left( \frac{y^2}{2} - 3y^2 \right) - \left( \frac{y^2}{2} + 3y^2 \right) \right) dy =$$

$$= \int_{-1}^0 -6y^2 dy = -6 \frac{y^3}{3} \Big|_{-1}^0 =$$

$$= -2(0 - (-1)^3) = -128.$$